

SALE, Morocco

WMO: 601350

Lat: 34.051N

Lon: 6.752W

Elev: 276

StdP: 14.55

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.6	42.6	33.7	28.6	53.3	37.1	32.8	50.0	22.9	58.3	19.3	58.7	3.0	140	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.4	91.1	69.9	86.0	69.8	82.5	69.9	75.3	82.6	73.6	80.4	72.4	78.9	9.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	124.4	79.6	71.5	117.7	77.5	70.1	111.8	76.0	38.8	82.8	37.3	80.8	36.2	78.8	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.0	16.3	14.2	DB	35.7	103.1	3.3	6.3	33.3	107.7	31.4	111.3	29.5	114.8	27.1	119.4
			WB	34.3	77.3	3.3	3.7	31.9	80.0	30.0	82.2	28.2	84.2	25.8	87.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.8	54.4	55.3	58.7	60.8	65.0	69.5	72.3	73.3	70.9	67.5	60.8	56.6	(6)
(7)		DBStd	7.79	3.79	4.01	4.62	4.07	4.77	4.53	4.13	3.62	3.53	4.16	4.68	3.95	(7)
(8)		HDD50	12	6	5	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD65	1431	329	273	205	138	54	7	0	0	1	17	145	262	(9)
(10)		CDD50	5057	142	152	271	325	467	584	693	723	628	544	323	205	(10)
(11)		CDD65	998	0	1	11	13	55	140	228	258	178	96	17	1	(11)
(12)		CDH74	5887	4	11	123	149	448	779	1331	1569	885	494	89	5	(12)
(13)		CDH80	1500	0	1	38	49	182	249	347	328	159	133	14	0	(13)
(14)	Wind	WSAvg	6.3	5.7	6.1	6.7	7.1	7.1	6.9	6.6	6.2	6.0	5.8	5.9	6.0	(14)
(15)	Precipitation	PrecAvg	22.2	3.1	2.5	2.3	2.2	0.7	0.2	0.0	0.3	1.9	3.5	4.6	(15)	
(16)		PrecMax	34.1	13.5	8.0	9.0	7.6	3.8	1.2	0.4	0.4	2.4	6.5	13.5	15.5	(16)
(17)		PrecMin	11.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	6.3	2.8	2.0	1.9	2.0	0.8	0.3	0.1	0.1	0.5	1.8	3.1	3.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.7	75.3	85.3	87.4	94.6	97.1	98.7	95.3	90.3	91.8	82.0	73.6	(19)
(20)			MCWB	60.3	58.9	64.9	65.8	67.6	70.9	75.7	70.6	72.4	64.8	65.2	59.9	(20)
(21)		2%	DB	66.7	69.6	76.6	77.4	84.5	86.0	87.0	86.5	84.1	82.8	76.0	69.5	(21)
(22)			MCWB	58.5	60.0	61.4	62.8	66.6	69.0	71.9	73.0	70.3	65.3	63.2	59.4	(22)
(23)		5%	DB	64.4	66.1	71.3	72.3	77.9	80.9	82.7	82.7	80.5	78.6	72.2	66.6	(23)
(24)			MCWB	58.1	59.1	60.8	62.4	65.5	69.1	72.3	72.3	70.0	66.7	63.4	59.4	(24)
(25)		10%	DB	62.4	63.8	67.7	69.5	73.7	77.3	79.8	80.5	78.5	75.6	69.7	64.7	(25)
(26)			MCWB	56.9	57.9	60.0	61.6	64.9	68.4	71.2	71.7	69.3	66.6	62.3	58.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.1	64.1	67.0	68.1	71.6	74.6	78.5	78.1	75.7	72.1	69.1	64.6	(27)
(28)			MCDWB	67.7	70.0	77.6	79.1	81.6	86.2	88.8	85.8	83.8	78.7	74.7	68.5	(28)
(29)		2%	WB	60.5	61.7	64.0	65.3	68.6	71.8	75.3	75.2	73.1	70.1	67.0	62.5	(29)
(30)			MCDWB	64.7	66.6	72.1	73.7	78.0	80.2	83.0	81.4	79.4	75.9	72.2	66.3	(30)
(31)		5%	WB	59.0	60.0	62.4	63.5	67.0	70.2	73.1	73.7	71.7	69.0	65.0	60.8	(31)
(32)			MCDWB	63.3	64.4	68.8	70.6	75.3	77.8	80.0	79.9	77.5	74.6	70.5	65.0	(32)
(33)		10%	WB	57.4	58.6	60.7	62.2	65.5	69.1	71.8	72.6	70.5	67.8	63.1	59.3	(33)
(34)			MCDWB	61.5	62.8	66.5	68.3	72.9	76.1	78.3	78.6	76.3	73.4	68.4	63.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.6	17.1	17.9	17.3	17.1	16.0	15.2	15.4	16.1	17.4	17.0	16.5	(35)
(36)			MCDBR	19.1	20.4	25.4	23.9	26.5	23.3	20.5	19.4	19.3	21.9	20.2	19.0	(36)
(37)			MCWBR	12.6	13.2	14.3	13.3	12.4	10.2	9.8	9.2	9.5	10.7	11.4	11.8	(37)
(38)		5% WB	MCDWB	16.5	18.0	20.4	20.2	20.2	18.1	17.6	16.1	16.5	17.2	17.2	15.4	(38)
(39)			MCWBR	11.7	12.8	13.2	12.6	11.1	9.2	9.5	9.0	9.5	10.3	11.2	10.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.330	0.357	0.409	0.405	0.429	0.461	0.498	0.493	0.441	0.403	0.363	0.338	(40)	
(41)		taud	2.509	2.420	2.251	2.293	2.249	2.157	2.033	2.065	2.244	2.362	2.455	2.505	(41)	
(42)		Ebn at Noon	279	282	274	280	273	263	252	252	260	262	264	269	(42)	
(43)		Edh at Noon	27	32	42	42	44	48	54	52	41	34	28	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg	933	1216	1590	1942	2203	2417	2429	2194	1791	1328	990	848	(44)	
(45)		RadStd	76	115	115	100	129	105	53	57	74	98	77	61	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) -2.00	(g) N/A	(h) -1.33	(i) -1.42	(j) N/A	(k) N/A	(l) N/A	(46)	
(47)	Regional (5 neighbors)	N/A	N/A	N/A	+1.82	N/A	N/A	N/A	N/A	N/A	+153	(47)

Nomenclature: See separate page